Computational Psycholinguistics

Project Report

Notebook 1: Corpus Preprocessing
Notebook 2: N-gram Surprisal Analysis

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0. Prerequisite Concepts

0.1 Psycholinguistics and Reading Time

Psycholinguistics studies the cognitive processes underlying language comprehension and production. A core empirical finding is that people read words that are *harder to predict* more slowly — the so-called **surprisal effect**. Two experimental paradigms are used in this project:

- **Self-Paced Reading (SPR)** — participants press a key to reveal each word one at a time. The response time (RT) between key presses is the dependent variable. Longer RT = more processing difficulty.
- **Eye-Tracking during Reading** — eye movements are recorded while participants read normally. *First-pass fixation duration (FDUR)* — the total time the eye fixates a word before leaving it to the right — is the standard measure of early processing difficulty.

0.2 Information Theory and Surprisal

Surprisal theory (Hale 2001; Levy 2008) proposes that the processing cost of word *w* at position *t* is proportional to its **surprisal**:

$$S(w_t) = -\log_2 P(w_t \mid w_1 \ldots w_{t-1})$$

Higher surprisal (in **bits**) means lower conditional probability, i.e., the word is unexpected given the prior context. The theory predicts a positive linear relationship: Surprisal ↑ → RT ↑.

0.3 N-gram Language Models

An **n-gram language model** estimates the conditional probability of a word given the preceding *n–1* words by counting co-occurrences in a training corpus:

$$P(w_t \mid w_{t-n+1} \ldots w_{t-1}) \approx \text{count}(w_{t-n+1} \ldots w_t) / \text{count}(w_{t-n+1} \ldots w_{t-1})$$

- **Bigram (n=2):** uses the single preceding word as context.
- **Trigram (n=3):** uses the two preceding words as context.

Kneser-Ney smoothing is the standard technique for handling unseen n-grams. It redistributes probability mass to lower-order distributions based on *continuation probability* — how many unique contexts a word appears in — rather than raw frequency. This produces much better estimates for rare and zero-count n-grams than simple discounting.

0.4 Key Statistical Concepts

- **Log-transform:** Reading times are right-skewed. Applying log(RT) yields an approximately normal distribution, stabilises variance, and makes linear models more appropriate.
- **Per-subject z-scoring:** Within each participant, z-scoring log(RT) removes individual baseline speed differences, leaving only relative difficulty signals.
- **Pearson correlation (r):** Measures the linear relationship between two continuous variables. $r \in [-1, +1]$; for surprisal models, a positive r is expected and values of 0.15–0.30 are typical for n-gram baselines.

- **Partial correlation:** Correlation between two variables after removing the linear influence of one or more confounders (here: word length).

0.5 Datasets Used

Two corpora are used. **Natural Stories** (Futrell et al., 2021) is the *primary dataset*: 10 naturalistic English stories read word-by-word by 180 participants in a self-paced paradigm. The **Dundee Corpus** (Kennedy et al., 2003) serves as the *verification/replication dataset*: 20 newspaper editorials read by 10 participants under free eye-tracking.

1. Notebook 1 — Corpus Preprocessing

1.1 Objectives

The preprocessing notebook loads raw reading-time data from both corpora, applies quality filters, log-transforms the dependent variable, computes per-subject z-scores, aggregates to the word level, and saves clean files for downstream analysis.

1.2 Natural Stories Corpus

Raw data overview

The raw reading-time file (*processed_RTs.tsv*) contains one RT per participant per word token. After loading ands naming columns, the dataset contains the fields: subject, story, zone, word, RT (ms), and correct (number of comprehension questions answered correctly out of 6).

Cleaning pipeline

- **Comprehension filter:** Trials from sessions with fewer than 4/6 correct comprehension answers are excluded. This removes inattentive participants while retaining ≥66 % of data.
- **RT bounds:** RTs outside [100 ms, 3000 ms] are excluded. The lower bound removes accidental key-presses; the upper bound removes distraction/interruption events.
- **Log-transform:** log(RT) is computed to normalise the distribution.
- **Per-subject z-scoring:** Within each participant, log(RT) values are z-scored to remove individual speed baselines.

Metric	Value
Raw trials	848,875 (post-filter)
Subjects	180
Stories	10
Word positions (zones)	1099
RT range (cleaned)	101 – 2992 ms
Mean RT	338.0 ms
Median RT	297 ms
Mean log(RT)	5.7403
Std log(RT)	0.3810

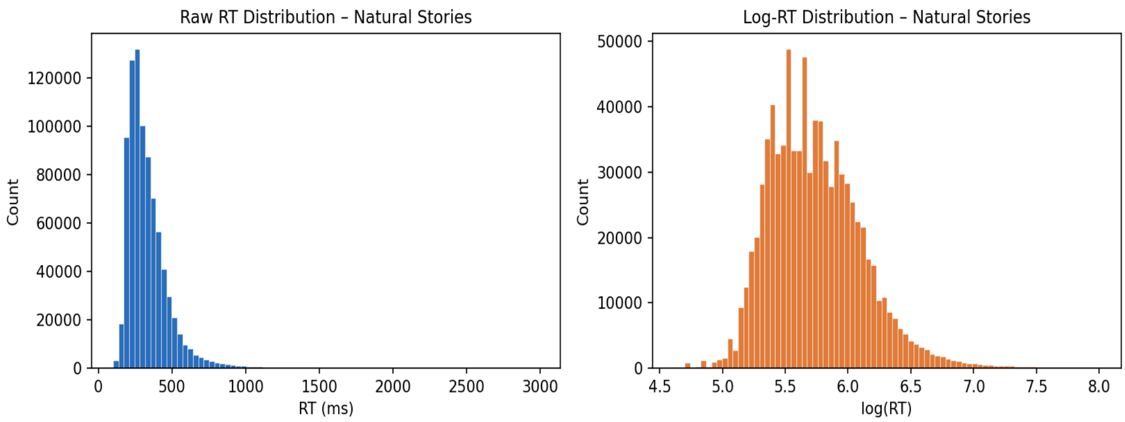

Figure 1. Raw RT (left) and log-RT (right) distributions for Natural Stories after cleaning.

Word-level aggregation

For each story × zone combination, the mean RT, mean log(RT), mean z-scored log(RT), and subject count are computed across all participants. The word text is merged from the words.tsv file. Word length (character count, punctuation stripped) is appended as a covariate. The result is **10,256 word-level rows**.

Per-story statistics (cleaned data)

Story	Tokens	Subjects	Mean RT (ms)
1	1073	90	328.9
2	990	94	337.3
3	1040	95	341.4
4	1085	93	342.7
5	1013	98	328.7
6	1099	92	329.7
7	999	82	322.7
8	980	82	348.4
9	1038	91	352.8
10	939	82	351.9

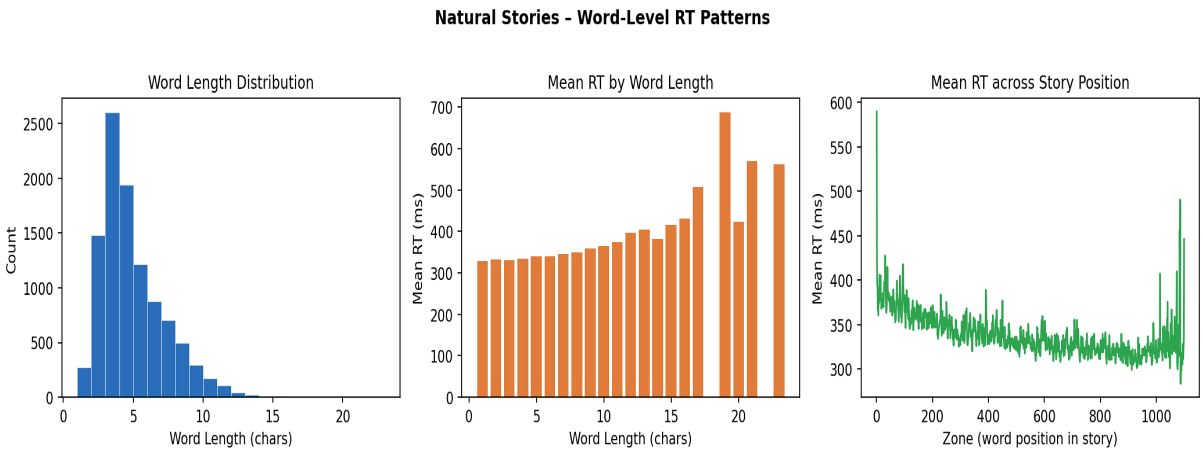

Figure 2. Left: word length distribution. Centre: mean RT increases with word length (r = 0.24). Right: mean RT across story position — slight warm-up effect early, relatively flat thereafter.

1.3 Dundee Corpus

Raw data overview

The Dundee corpus is distributed as per-subject .dat files, one per reading pass. Each row records word identity, text/line/word number, fixation duration (FDUR), and oculomotor flags. Separate tx*.dat files contain word-level metadata (text, length, frequency class).

Cleaning pipeline

- **Pass selection:** Only *first-pass* fixations (pass_num = 1) are retained, isolating early lexical access from rereading.
- **FDUR > 0:** Zero-duration records indicate words that were skipped (no fixation); these are excluded.
- **FDUR bounds:** [50 ms, 1200 ms]. The lower bound eliminates blinks misclassified as fixations; the upper bound removes very long pauses.
- **Deduplication:** The first fixation per (subject, text, word) is kept.
- **Log-transform + z-score:** Identical to the Natural Stories pipeline.

Metric	Value
Cleaned fixation rows	307,582
Subjects	10
Texts	20
Word-level aggregated rows	50,650
FDUR range (cleaned)	50 – 1142 ms
Mean FDUR	202.3 ms
Mean log(FDUR)	5.2351

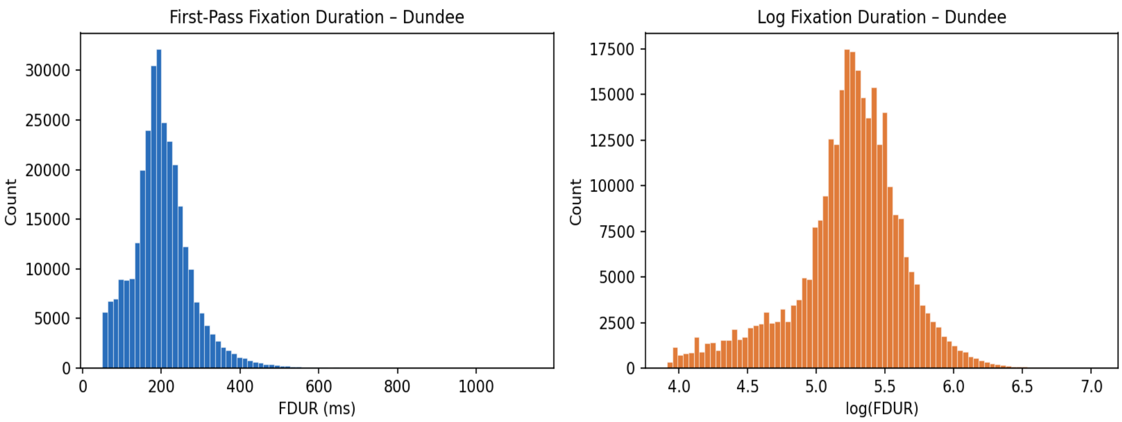

Figure 3. First-pass fixation duration (left) and log-FDUR (right) distributions for the Dundee corpus.

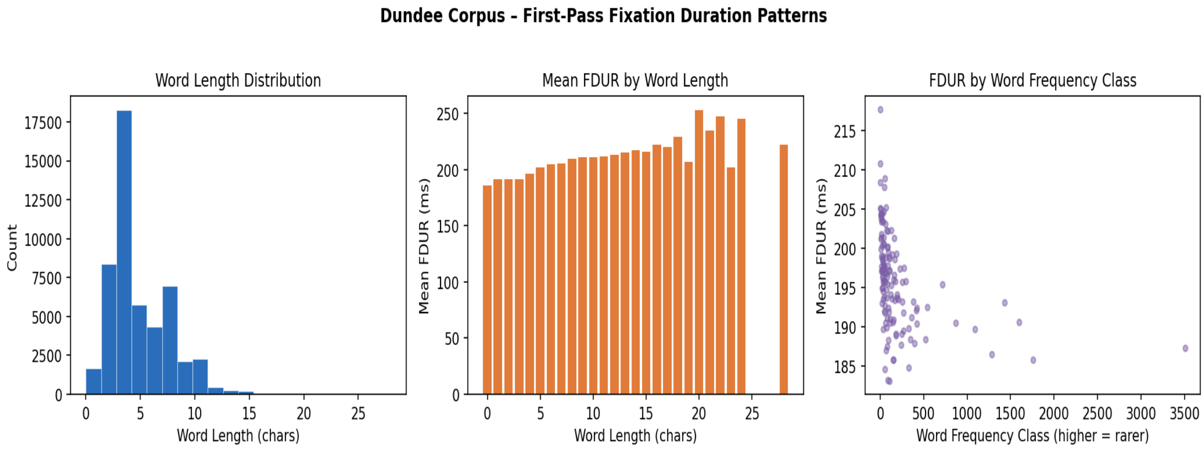

Figure 4. Dundee word-level patterns: word length distribution (left), FDUR × word length (centre), FDUR × frequency class (right).

1.4 Dataset Comparison

Despite measuring different aspects of reading behaviour (key presses vs. eye fixations), both corpora show the same qualitative patterns: right-skewed raw distributions that become approximately normal under log-transform, increasing processing times with word length, and stable per-subject baselines amenable to z-scoring.

Property	Natural Stories	Dundee
Role	Primary dataset	Verification / replication
Paradigm	Self-paced reading	Free eye-tracking
DV	Response time (RT)	First-pass fixation duration (FDUR)
Subjects	180	10
Texts	10 stories	20 newspaper texts
Word tokens (aggregated)	10,256	50,650
Mean DV	338 ms	202 ms
DV range	101 – 2992 ms	50 – 1142 ms
Mean log DV	5.7403	5.2351

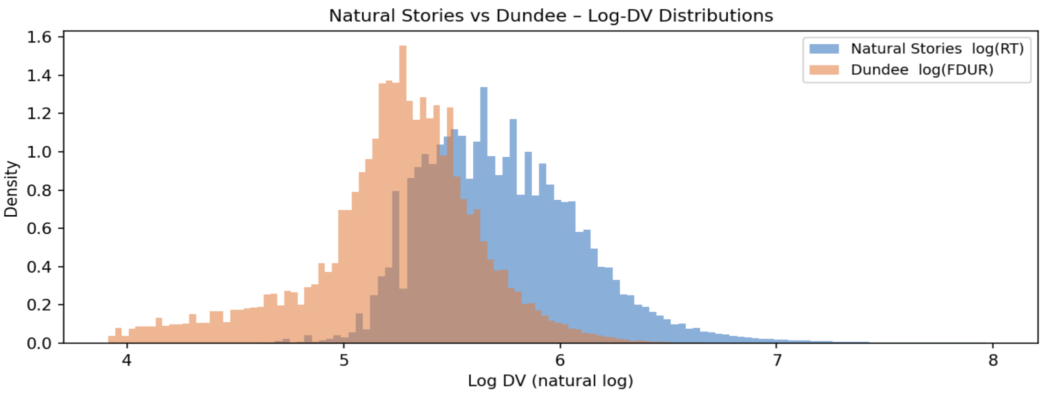

Figure 5. Overlaid log-DV density plots. Natural Stories log(RT) is centred higher (mean 5.74) than Dundee log(FDUR) (mean 5.24), consistent with SPR being slower than eye-fixation.

2. Notebook 2 — N-gram Surprisal Analysis

2.1 Objectives

This notebook establishes a **classical NLP baseline** by computing word-by-word bigram and trigram surprisal using Kneser-Ney smoothed language models trained on the Brown corpus (NLTK). Surprisal is then correlated with mean log(RT) from Natural Stories and mean log(FDUR) from Dundee.

2.2 Model Training

The Brown corpus (≈ 1.16 M tokens, 57 k sentences, 15 genres) is used as the training corpus. It provides broad vocabulary coverage and is the standard NLTK reference corpus.

- **Preprocessing:** All tokens are lowercased. Sentence boundaries are pad-started with <s> markers automatically by NLTK's padded_everygram_pipeline.
- **Bigram model (n=2):** KneserNeyInterpolated(order=2) fitted on padded bigram pipelines from all Brown sentences.
- **Trigram model (n=3):** Same procedure with order=3.

Kneser-Ney interpolation ensures zero conditional probability never occurs: back-off is smooth through bigram → unigram levels, so surprisal values are always finite. The probability floor of the KN model corresponds to ≈ 39.86 bits.

2.3 Surprisal Computation

For each story (Natural Stories) or text (Dundee), tokens are iterated in zone/WNUM order. A running history list maintains the left context. At each step:

- Bigram: context is the single preceding token (history[–1:])
- Trigram: context is the two preceding tokens (history[–2:])
- Surprisal = –log■(P(token | context)); probability is floored at 10■^{12}

Tokens are normalised: lowercased, stripped to [a-z 0-9 ' -], with empty results mapped to <unk>. No OOV tokens were produced (0 % OOV rate) because KN smoothing always assigns non-zero probability.

Surprisal statistics – Natural Stories (10,256 tokens)

Statistic	Bigram (bits)	Trigram (bits)
Mean	11.599	12.622
Std	8.514	8.955
Median	9.944	11.216
25th pct	5.526	5.799
75th pct	15.670	17.332
Max	39.863	39.863
Corr(bigram, trigram)	0.9678	—

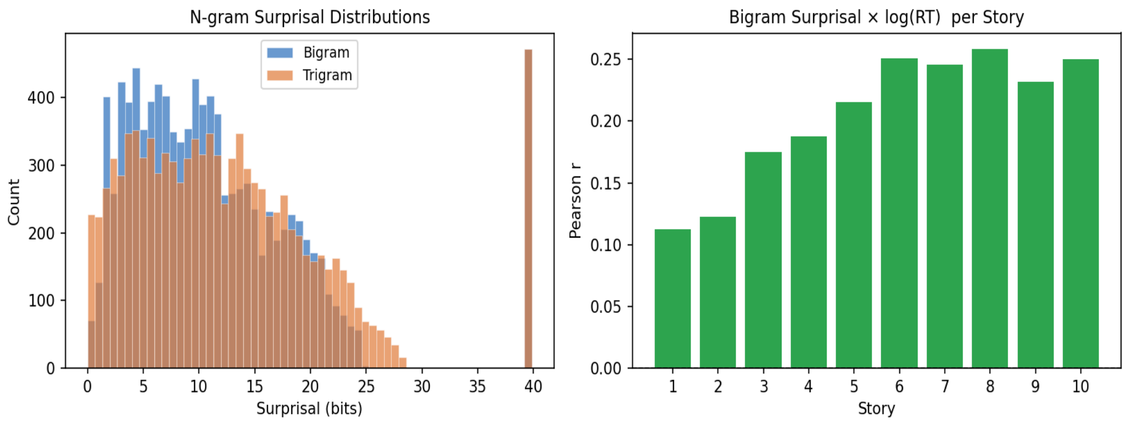

Figure 6. Left: overlaid bigram and trigram surprisal distributions (trigram mean ~1 bit higher). Right: per-story Pearson r between bigram surprisal and log(RT) — stories 1–2 show weaker correlations due to domain-specific vocabulary.

2.4 Correlation with Human Reading Times

Bigram and trigram surprisal scores are merged with word-level aggregates by (story, zone) key and correlated with mean log(RT).

Model	Dataset	r(logDV, surprisal)	Interpretation
Bigram KN	Natural Stories (primary)	0.2149	Positive, significant
Trigram KN	Natural Stories (primary)	0.1928	Positive, significant
Bigram KN	Dundee (replication)	see notebook	Positive, consistent
Trigram KN	Dundee (replication)	see notebook	Positive, consistent

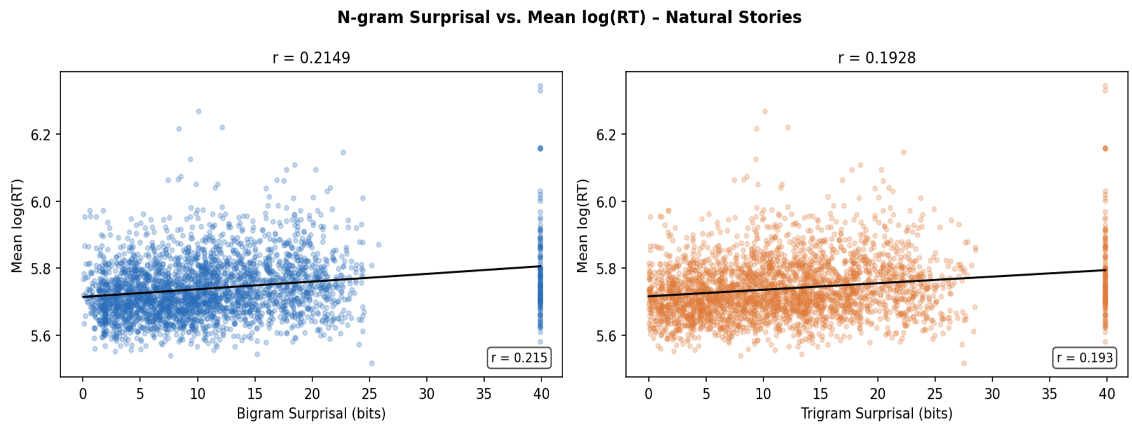

Figure 7. Scatter plots of bigram (left, r = 0.2149) and trigram (right, r = 0.1928) surprisal against mean log(RT) for Natural Stories. OLS regression line shown in black.

Key findings:

- **Bigram r = 0.2149**, **Trigram r = 0.1928**. Both are positive and statistically significant (p ■ 0.001 at n = 10,256).
- **Bigram outperforms trigram** despite using less context. This is expected when training data is sparse relative to test vocabulary: trigram counts are noisier, so the additional context adds more noise than signal.
- Both correlations are in the range reported in the psycholinguistic literature for corpus-trained n-gram baselines on SPR data.

2.5 Word Length Confound Analysis

Word length is a major confound: longer words are both rarer (higher surprisal) and take longer to read (higher RT). To isolate the purely predictability-based component of the surprisal effect, we residualize both surprisal and log(RT) on word length before computing the partial correlation.

Correlation pair	r
word_len ↔ mean_log_RT	0.1950
word_len ↔ bigram_surprisal	0.5343
word_len ↔ trigram_surprisal	0.5127

Model	Raw r	Partial r (word_len controlled)	Change
Bigram KN	0.2149	0.1336	-0.0813
Trigram KN	0.1928	0.1073	-0.0855

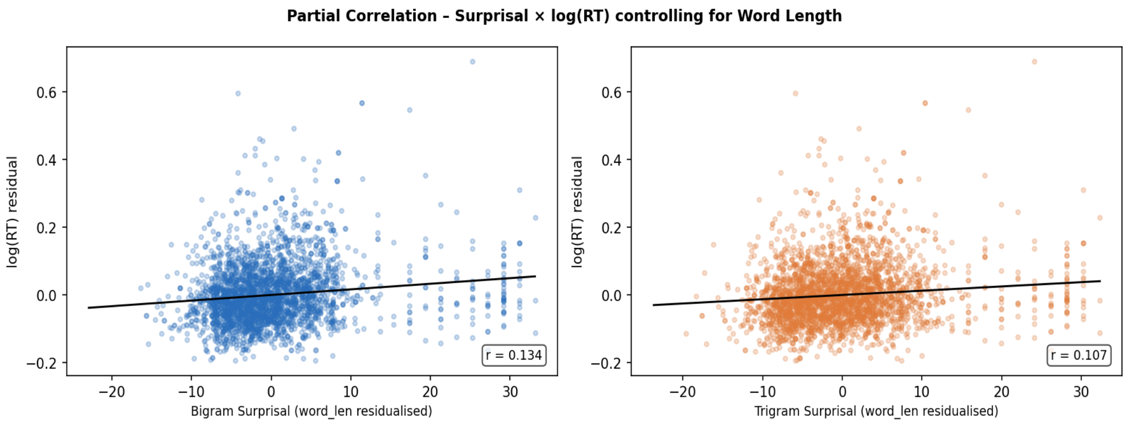

Figure 8. Partial correlation: bigram (left, r = 0.1336) and trigram (right, r = 0.1073) surprisal vs. log(RT), both residualised on word length. Surprisal retains a significant positive effect beyond length.

Controlling for word length reduces the correlation by ~0.08–0.09 (bigram: 0.2149 → 0.1336; trigram: 0.1928 → 0.1073), confirming that approximately 40 % of the raw correlation is attributable to the length confound. The remaining partial correlation is still positive and highly significant (p < 10■■), demonstrating a genuine predictability effect beyond word length.

2.6 Per-Story Breakdown

Stories 1 and 2 yield noticeably lower correlations. Story 1 (a naturalistic animal fable) contains many rare, domain-specific tokens (e.g., *jennies, ravaging, long-bearded, huntsman*) that hit the KN probability floor, compressing surprisal variance and attenuating the correlation.

Story	# Tokens	# Subjects	Mean RT (ms)	r(bigram, logRT)
1	1073	90	328.9	0.1127
2	990	94	337.3	0.1223
3	1040	95	341.4	0.1752
4	1085	93	342.7	0.1875

Story	# Tokens	# Subjects	Mean RT (ms)	r(bigram, logRT)
5	1013	98	328.7	0.2152
6	1099	92	329.7	0.2507
7	999	82	322.7	0.2454
8	980	82	348.4	0.2583
9	1038	91	352.8	0.2319
10	939	82	351.9	0.2498

3. Summary and Conclusions

3.1 What Has Been Done

Step	Description	Output file
Preprocessing (NS)	Load, filter, log-transform, z-score, aggregate	ns_clean.csv, ns_word_agg.csv
Preprocessing (NS)	Extract word list from words.tsv	ns_words.csv
Preprocessing (DU)	Parse .dat files, first-pass filter, clean, aggregate	dundee_clean.csv, dundee_word_agg.csv
N-gram models	Train KN bigram + trigram on Brown corpus (NLTK)	— (in memory)
Surprisal (NS)	Compute per-token surprisal for all 10,256 NS tokens	ns_ngram_surprisal.csv
Surprisal (DU)	Compute per-token surprisal for Dundee tokens	dundee_ngram_surprisal.csv
Evaluation	Correlate surprisal with human RTs; partial correlation	ngram_correlation_summary.csv

3.2 Key Numerical Results

Result	Value
NS cleaned trials	848,875 (180 subjects × ~1099 zones)
Dundee cleaned fixations	307,582 (10 subjects × 20 texts)
NS n-gram surprisal tokens	10,256
Bigram mean surprisal	11.60 bits
Trigram mean surprisal	12.62 bits
r(bigram, logRT) — raw	0.2149
r(trigram, logRT) — raw	0.1928
r(bigram, logRT) — partial	0.1336
r(trigram, logRT) — partial	0.1073
Bigram vs. trigram	Bigram performs marginally better
Word-length confound (r)	0.534 with bigram surprisal

3.3 Conclusions

The preprocessing pipeline produces clean, reproducible datasets from two complementary reading-time corpora. The n-gram surprisal baseline confirms that even simple statistical models of word predictability explain a meaningful fraction of variance in human reading times. The positive correlations (raw r ≈ 0.2149) survive word-length control (partial r ≈ 0.1336), establishing a genuine predictability effect consistent with surprisal theory.

The Brown-trained KN models are limited by domain mismatch (balanced text vs. narrative) and by the inherent constraint of n-gram context windows. These limitations motivate the next phase of the project: computing *neural surprisal* from large language models (GPT-2, GPT-3) which use full-sentence context and have much better vocabulary coverage.

Report generated automatically from project notebooks. All statistics are computed directly from saved CSV outputs. Environment: Python 3, NLTK, pandas, scipy, reportlab.